

Narrowband Channel Measurements and Statistical Characterization in Subway Tunnels at 1.8 and 5.8 GHz

Xuejian Zhang^{1b}, Ruisi He^{1b}, Senior Member, IEEE, Mi Yang^{1b}, Member, IEEE, Ziyi Qi^{1b}, Zhengyu Zhang^{1b}, Student Member, IEEE, Bo Ai^{1b}, Fellow, IEEE, and Ruifeng Chen^{1b}

Abstract—Railway wireless communications play an increasingly significant role in train control and signal transmission. Communication-based train control system (CBTC) can improve the safety, efficiency and reliability of train operations, and is widely used in subway wireless communication system. Therefore, radio propagation characterization and channel modeling in subway tunnels for CBTC are significant for designing reliable wireless communication system. This paper presents extensive narrowband propagation measurements conducted in a subway tunnel at 1.8 GHz and 5.8 GHz with different scenarios, which are straight tunnel scenario, station scenario and curved tunnel scenarios. Based on the measurements, some typical large and small-scale channel parameters are extracted and modeled, such as path loss, shadow fading and channel correlation, fading depth, level crossing rate, average fade duration and envelope distribution of fading. Effects of frequencies, polarizations, tunnel scenarios on wireless propagation are discussed in detail. Generally, higher frequency, vertical polarization, station and tunnel curvature will bring additional propagation losses and larger fluctuations of fading. Moreover, it is found that envelope distribution of small-scale fading is fitted well by Nakagami distribution, and parameter- m is proportional to distance. Results can support realization of subway intelligent transportation system.

Index Terms—Wireless communication, channel measurement, large-scale fading, small-scale fading, subway tunnel.

I. INTRODUCTION

RECENTLY, low-carbon travel and low-carbon life have become behaviors respected by the whole society, urban railway transit is widely regarded as a low-carbon and environmentally friendly travel method that saves energy and reduces emissions [1]. In addition, with rapid development of intelligent urban railway transit and extensive application of CBTC, subway has gradually become an ideal travel choice for people due to its advantages of safety, efficiency, and green energy [2]. However, ever-increasing subway passenger flow also poses a major challenge to safe operation of subway [3]. Among them, providing reliable wireless communication, monitoring train status, improving operating efficiency and ensuring personnel safety during subway operation have become key issues that need to be paid attention to in urban subway systems [4]. Channel modeling provides necessary guarantee for optimal design and safe operation of subway communication systems [5], [6], [7]. To meet the above requirements and fully consider design, operation and optimization of subway system, a comprehensive understanding of channel characteristics and accurate tunnel channel models are essential [8], [9], [10].

At present, propagation characteristics of radio waves in tunnels have been widely investigated. Related work on channel modeling in tunnel is listed in Table I. Statistical methods and deterministic methods are main research methods [11], [12]. Deterministic methods mainly include: modal analysis methods based on waveguide theory [13], [14], [15], methods based on Geometric Optics [16], numerical methods for solving closed-form solutions of Maxwell's equations [17], [18], [19], and methods based on Ray Tracing (RT) [20], [21], [22], [23]. Deterministic methods rely on a fine description of propagation environment, and can reproduce propagation law of electromagnetic waves in tunnel relatively perfectly, and excessive computation and complete mastery of environmental information are also limitations. Statistical methods do not require detailed dissemination of environment, and amount of calculation is relatively small. Field measurements in tunnels are required to obtain channel data [24]. Therefore, statistical methods are mainly based on empirical statistical models of measurement

Manuscript received 3 August 2023; revised 7 November 2023, 1 January 2024, and 6 February 2024; accepted 18 February 2024. Date of publication 22 February 2024; date of current version 16 July 2024. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant 2022JBQY004, in part by the National Natural Science Foundation of China under Grant 62271037, Grant 62221001, Grant 62301022, and Grant U21A20445, in part by the State Key Laboratory of Advanced Rail Autonomous Operation under Grant RCS2022ZZ004, in part by Young Elite Scientists Sponsorship Program by CAST under Grant 2022QNRC001, in part by the Science and Technology Project of China Academy of Railway Sciences under Grant 2022YJ033 and Grant 2023YJ074, and in part by the Program for Science & Technology R&D Plan Joint Fund of Henan Province under Grant 225200810112. The review of this article was coordinated by Prof. Vittorio Degli-Esposti. (Corresponding authors: Ruisi He; Mi Yang.)

Xuejian Zhang, Ruisi He, Ziyi Qi, Zhengyu Zhang, and Bo Ai are with the School of Electronics and Information Engineering and the Frontiers Science Center for Smart High-speed Railway System, Beijing Jiaotong University, Beijing 100044, China (e-mail: 21120172@bjtu.edu.cn; ruisi.he@bjtu.edu.cn; qzy971031@gmail.com; 21111040@bjtu.edu.cn; aibo@ieee.org).

Mi Yang is with the School of Electronics and Information Engineering and the Frontiers Science Center for Smart High-speed Railway System, Beijing Jiaotong University, Beijing 100044, China, and also with the Henan High-Speed Railway Operation and Maintenance Engineering Research Center, Zhengzhou 451460, China (e-mail: myang@bjtu.edu.cn).

Ruifeng Chen is with the Institute of Computing Technology, China Academy of Railway Sciences Corporation Ltd., Beijing 100081, China (e-mail: ruifeng_chen@126.com).

Digital Object Identifier 10.1109/TVT.2024.3368548

TABLE I
RELATED WORK ON CHANNEL MODELING IN TUNNEL

Ref.	Freq. [MHz]	Method.	Meas. for Validation	Scenarios	Tunnel Dimension (W×H×L) [m]	Pol.	Parameters	Modeling
[14]	900, 1800, 2448	Modal Theory	Narrowband	Straight subway tunnel, etc	Rectangular 3.8×4.3×110, etc	HH VV	PL	Waveguide model
[15]	433, 868, 1800, 2400	Modal Theory	Narrowband	Straight railway tunnel	Horseshoe-shaped 4.65×5×7700	HH	RSSI	Waveguide model
[16]	1000	Geometric Optics	-	Straight tunnel	Rectangular 4.62×2.31×2000	HH VV	Electric field	Geometrical optics approach
[17]	2400	Numerical Method	-	Straight, curved simulation tunnel	Semicircular 8.3×5.8×500	-	PL	FDTD
[19]	900, 2400	Numerical Method	-	Straight, curved simulation tunnel	Rectangular 8.5×5.3×500	VV	RSSI	VPE
[21]	1400, 40000	RT	Wideband	Straight experimental tunnel	Rectangular 5×3×50 Horseshoe-shaped 4×3×50	VV HH HV VH	PL, RMS DS	Simulation analysis
[22]	1800, 5800	RT	-	Straight simulation tunnels	Rectangular 6×6×500, etc	VV	PL, K-factor, RMS DS, AS	Simulation analysis
[23]	30000 to 100000	RT	-	Curved simulation tunnel	Rectangular 8×5×400	-	PL, RMS DS	Simulation analysis
[25]	2400	Channel Measurement	Narrowband	Straight, curved subway tunnel	Arched 8×6×500	VV	PL	Two-Slope model
[26]	400	Channel Measurement	Narrowband	Straight railway tunnel,etc	Arched 4.7×4.5×520, etc	VV	PL	Four-Slope model
[28]	2800 to 5000	Channel Measurement	Wideband	Straight railway tunnel	Semicircular 8.6×6.1×500	-	PL, K-factor, Fading Distribution	Statistical analysis
[33]	900	Modal Theory and RT	Narrowband	Curved railway tunnel	Arched 10.7×5.4×4000	VV	RSSI	Combined wave- and RT
[34]	465, 820	Channel Measurement	Narrowband	Straight railway tunnel	Arched 5.8×4×980	VV	PL	Statistical analysis
[35]	1350	Channel Measurement	Wideband	Straight road tunnel	Arched 9.7×4.63×590	VV HH HV VH	PL, CPR, XPD Fading Distribution RMS DS	Statistical analysis
[37]	455, 915, 2450, 5800	Modal Theory and RT	Narrowband	Straight railway tunnel	Arched 5.8×4×980	VV HH	RSSI	Combined wave- guide and RT
[40]	900, 1800	Channel Measurement	Narrowband	Straight subway tunnel, etc	Arched 3.43×2.6×286, etc	HH VV	PL, SF	Statistical analysis
[42]	1890	Channel Measurement	Narrowband	Straight road tunnel	Arched 10×5×1130	-	PL, SF, LCR, AFD, Fading Distribution	Statistical analysis
[44]	3500, 5600	Modal Theory and RT	Narrowband	Straight, curved cascaded tunnel	Arched 2.78×3.4×650	-	PL	Combined wave- guide and RT
[48]	920, 2400, 5705	Channel Measurement	Narrowband	Curved subway tunnels	Arched 7.59×5.52×500, etc	VV HH	PL, SF, Channel correlation	Statistical analysis
[50]	2400	Channel Measurement	Narrowband	Curved subway tunnels	Arched 9.8×6.2×1863, etc	VV HH	PL, SF, LCR, AFD Fading Distribution	Statistical analysis

* W: Width; H: Height; L: Length; FDTD: Finite Difference Time Domain; VPE: Vector Parabolic Equation; RSSI: Received Signal Strength Indicator; RMS DS: Root Mean Square Delay Spread; AS: Angle Spread; CPR: Co-Polar Ratio; XPD: Cross-Polar Discrimination.

data, such as double-slope models [25], four-slope models [26], and five-region models [27]. However, various methods and models mentioned above are mainly aimed at large-scale propagation characteristics such as path loss (PL) and shadow fading (SF) in tunnels which are crucial in network deployment and optimization, and have almost no description of small-scale propagation characteristics which also play an significant role in error control coding, interleaving, and equalization algorithms [8], such as fading depth (FD), level crossing rate (LCR), average fade duration (AFD) and statistical distributions. In [28], both large and small-scale fading characteristics deduced from measurement data in semicircular tunnel are presented. In the case of small-scale fading, a Rice distribution can fit measurement well, and a uniform distribution can match the phase of the electric field. A measurement was carried out in a subway tunnel and some propagation characteristics, such as LCR and AFD, have been computed and discussed in [29]. In future tunnel channel measurements, more channel statistics related to small-scale fading in subway tunnel scenarios are needed. Therefore, accurate channel models that fully consider both large and small-scale fading are still rarely studied.

On the other hand, current research on various tunnel scenarios is also extensive. For application scenario, there are subway tunnels [30], [31], [32], railway tunnels [33], [34], road tunnels [29], [35] and underground mine tunnels [36], [37], [38]. For cross-section of tunnel, it can be divided into rectangular tunnels [39], [40], arched tunnels [26], [35], [41], circular tunnels [42], etc. For tunnel strike, there are straight tunnels [15], [39], curved tunnels [23], [43], straight and curved cascaded tunnels [44]. Different from other tunnels or open environments, subway tunnels are underground and is almost completely closed, dimension of which is also narrower [8]. More importantly, waveguide effect in tunnel is observed obviously [45]. Various leaky cables, distributed antennas and intelligent detection equipment need to be laid on the inner wall of tunnels, and the distance between base stations is smaller, often only 100–500 m, thus it is difficult to carry out field measurements. It is also for above reasons that research on wireless propagation characteristics in subway tunnels mostly relies on simulation using ray tracing [46], [47]. There are few works on actual channel measurements in realistic subway tunnels. Moreover, propagation scenarios of subway tunnels involved in previous studies are not comprehensive, influence of stations and tunnel curvature in subway tunnels on large and small-scale propagation characteristics are usually ignored. Furthermore, existing work only considers limited frequency bands in channel measurement. Frequencies widely used in CBTC in the world mainly include 900 MHz, 1800 MHz, 2.4 GHz and 5.8 GHz [48]. In the past few decades, frequencies such as 900 MHz, 1800 MHz, and 2.4 GHz have been extensively studied [25], [49]. However, as a hotspot frequency of CBTC, there are few related researches on 5.8 GHz in subway tunnels, and certain significant characteristics of tunnel channels have not been investigated sufficiently.

In summary, current channel modeling in tunnels mostly focuses on large-scale propagation characteristics, and there are few researches on small-scale propagation characteristics. Due

to special environment of subway tunnels, channel modeling of diversified tunnel scenarios in the same tunnel is insufficient, and influence of straight tunnel, curved tunnel and station scenarios on large and small-scale wireless propagation characteristics has not been thoroughly studied. Finally, frequencies of channel measurement in subway tunnels are not complete, statistical modeling of 5.8 GHz as an important frequency band is still limited.

To solve the above problems, we carry out a large number of narrowband channel measurements in realistic subway tunnels, and sufficient channel measurement data is obtained. On this basis, this paper discusses in detail wireless propagation characteristics of two significant frequencies, 1.8 GHz and 5.8 GHz, in four different subway tunnel scenarios, fully considering influence of frequency, polarization, tunnel curvature and stations on propagation. Propagation characteristic parameters of large and small-scale fading are extracted and comprehensive analysis is carried out. The results of this study are expected to provide guidance for link budget, handover scheme design, and network planning for intelligent urban railway systems.

The remain of the paper is organized as follows. Measurement campaigns are presented in Section II. In Section III, large-scale propagation characteristic parameters of 1.8 GHz and 5.8 GHz in different tunnel scenarios are extracted and analyzed. Section IV draws conclusions.

II. MEASUREMENT CAMPAIGN

A. Measurement System

Extensive narrowband channel measurements are carried out in a realistic tunnel in Beijing. Measurement environment is shown in Fig. 1(a). Cross-section of subway tunnel is a horseshoe shape, which is widely used. Width and height of tunnel are 5.4 m and 4.8 m respectively and subway train is 94 m length, 3 m width and 3.5 m height.

Equipment used in measurements is based on Enhanced Ultra-High Throughput transmission standard developed by NUFONT [51], which has been commercialized and widely used in a number of subway lines in China, providing necessary security for subway operation. The measurement system consists of transmitter (Tx) and receiver (Rx). Frequency bands used in measurements are 1.8 GHz and 5.8 GHz. Measurement systems of these two frequencies are independent of each other. Detailed parameter configuration of measurement system can be found in Table II.

- i) Transmitting system: transmitting system is composed of base transceiver stations (BTSs) and transmitting antennas, as shown in Fig. 1(a). Transmitting antennas of 1.8 GHz and 5.8 GHz are connected to the corresponding base stations through radio frequency cables respectively, and installed on inner side of tunnel wall. It should be noted that a BTS is connected to two transmitting antennas at the same time. Two antennas have the same configurations and are used to cover front and rear of tunnel, respectively. Field strength signals in measurements all come from transmitting antennas on the same side, which are not interfered by signals from the other side. BTSs used

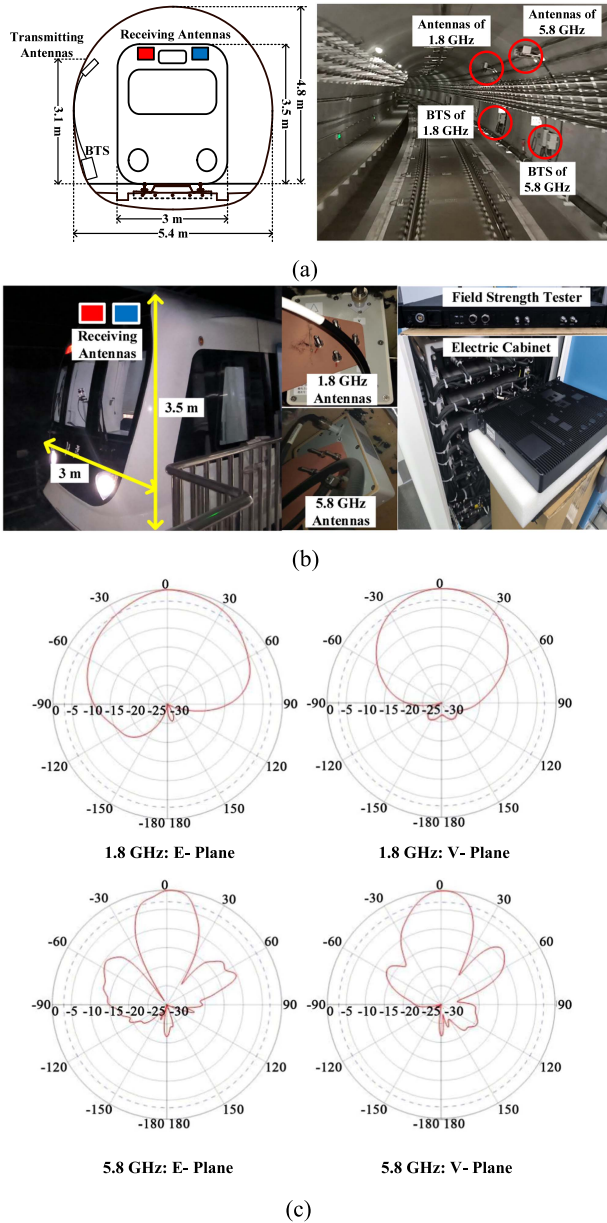


Fig. 1. (a) Front view of measurement tunnel. (b) Photos of subway train and antennas. (c) Antenna patterns of 1.8 GHz and 5.8 GHz antennas.

to transmit 1.8 GHz and 5.8 GHz signals are NPEC02S-04 and NPEC01S04, respectively. Transmitting power of BTSs is 12–18 dBm for different BTSs, and bandwidths corresponding to 1.8 GHz and 5.8 GHz are 20 MHz and 80 MHz, respectively. Besides, we use two kinds of antennas, which are horizontally polarized (H-Pol) and vertically polarized (V-Pol). Antennas with the same frequency and different polarizations are installed in the same antenna pod. That is to say, both transmitting and receiving side exist horizontally and vertically polarized antennas for 1.8 GHz and 5.8 GHz to transmit or receive signals at the same time, which work independently and are free from mutual interference. As shown in Fig. 1(a) and (c), for 1.8 GHz, we apply NPEA01S-VH-1800 panel

TABLE II
DETAILED PARAMETERS OF THE MEASUREMENT SYSTEM

Frequency	1.8 GHz	5.8 GHz
Transmitting power of BTSs	12–18 dBm	
Transmission bandwidth	20 MHz	80 MHz
Tx/Rx antenna gain	11 dBi	17 dBi
Polarization	Horizontal/Vertical	
Height of Tx	3.1 m	
Height of Rx	3.5 m	
System loss	17 dB	12 dB
Antenna radiation pattern	E: 80° V: 45°	E: 32° V: 32°
Isolation	≥ 20 dB	≥ 25 dB
Front-to-back ratio	≥ 25 dB	
Sampling rate	500 snapshots/s	1000 snapshots/s
Train speed	40 km/h	

antennas, of which operating frequency band is 1785–1805 MHz, antenna gain is 11 dBi, horizontal and vertical beamwidths are 80° and 45°, respectively. Front-to-back ratio and port isolation are not lower than 20 dB and 25 dB, respectively. Similarly, NPEA01S-4X4-5158 panel antennas are used for 5.8 GHz measurements with gain of 17 dBi and working frequency of 5150–5850 MHz. Horizontal and vertical beam widths are both 32°, and front-to-back ratio and port isolation are higher than 25 dB. All antennas in measurement system have been tested in an anechoic chamber to calibrate antenna gain and radiation patterns.

- ii) Receiving system: receiving system consists of receiving antennas and field strength testers, as shown in Fig. 1(b). Receiving antennas are fixed at roof of train and is approximately 3.5 m height (i.e., the height of train). The red and blue rectangles in Fig. 1(b) represent panel antennas for 1.8 GHz and 5.8 GHz, respectively. Configuration parameters of receiving antennas are exactly the same as those of transmitting antennas. Receiving antennas are connected to the on-board electrical cabinet, and the latter is connected to field strength testers through radio frequency cables. Therefore, reception of wireless field strength signal can be realized. NPES02S-04 and NPES01S-04 vehicle-mounted field strength testers are used to receive 1.8 GHz and 5.8 GHz signals respectively, of which threshold level is -90 dBm. Field strength testers include a distance sensing module, which can record and store distance between transmitting and receiving antennas. It is noteworthy that NPETO1S-03 synchronous timing unit is deployed on both Tx and Rx to ensure time synchronization.

B. Measurement Environment and Data Processing

Track of measurements is demonstrated in Fig. 2. Subway train keeps moving at a constant speed of 40 km/h during measurements, passing through three stations and four base

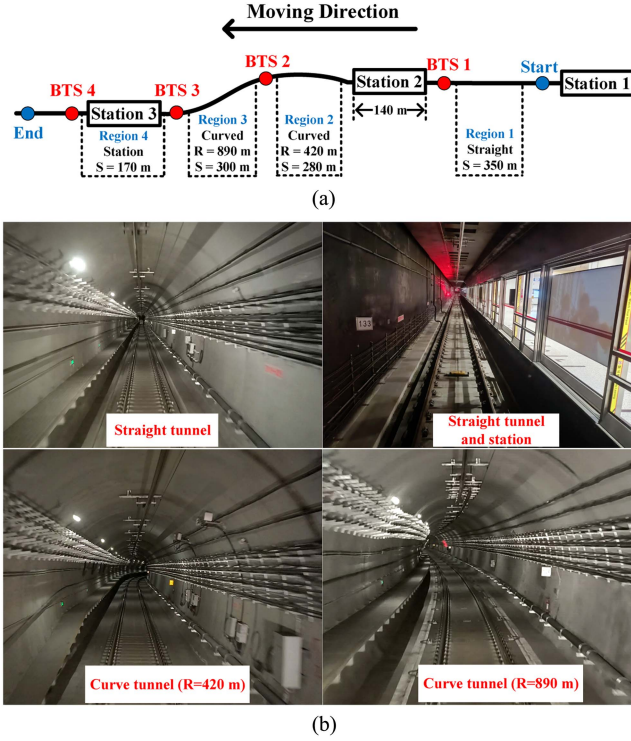


Fig. 2. (a) Skeleton map of the tunnel. (b) Photos of four cases.

stations. There exists four tunnel scenarios along the track as shown in Fig. 2. Radius of curvature and length of tunnel are represented by R and S , respectively. The basic situations of four tunnel scenarios are as follows:

- 1) Straight tunnel scenario: distributed in Region 1, receiving wireless signals from BTS 1, with a total length of nearly 350 m. There is a strong line of sight (LOS) in the first Fresnel region between Tx and Rx.
- 2) Straight tunnel and station scenario: distributed in Region 4, radio signals are provided by BTS 4, nearly 170 m, of which station is 140 m long and straight tunnel part is 30 m long. Similarly, due to shorter distance between transceivers, there is still a strong LOS if antenna bins are ignored. Moreover, surface material of station is quite disparate from tunnel walls, which will have impacts on radio propagation compared with straight tunnel.
- 3) Curved tunnel scenario ($R = 420$ m): distributed in Region 2, covered by BTS 2, nearly 280 m long and has a radius of curvature of 420 m. There exists no LOS, and curved tunnel brings more reflection, refraction and scattering to propagation, resulting in additional propagation losses.
- 4) Curved tunnel scenario ($R = 890$ m): distributed in Region 3, covered by BTS 3, total length of nearly 300 m and radius of curvature is 890 m. A larger curvature radius means that curvature of tunnel is smaller, and there is still no LOS, which will bring additional propagation losses.

In addition, subway train repeatedly travels on the same route five times, which means that channel measurements are conducted five times for each tunnel scenarios, and we obtain sufficient measurement data for channel characterization. Sampling rates of 1.8 GHz and 5.8 GHz are 500 snapshots/s and

1000 snapshots/s, i.e., every 0.13 and 0.21 wavelengths, one snapshot of 1.8 GHz and 5.8 GHz field strength signals can be sampled, respectively. With such high sampling rates, large and small-scale fading characteristics of propagation in subway tunnels can be obtained. Using a sliding average window of 40 wavelengths on distance scale can eliminate small-scale fading, and thus large-scale components can be extracted [52]. Furthermore, small-scale components can be obtained by subtracting large-scale components from original channel fading.

III. RESULTS AND ANALYSIS OF LARGE-SCALE COMPONENTS

A. Path Loss and Shadow Fading

After removing small-scale fading by a sliding average window of 40 wavelengths, the measured large scale propagation loss $L(d)$ can be obtained as follows [53]:

$$L(d) = P_t + G - P_r(d) - L_{\text{system}} \quad (1)$$

where P_r and P_t are powers of received and transmitted signals, respectively, d is distance between Tx and Rx, G is total gain from transmitting and receiving antennas, and L_{system} is total attenuation in measurement system, such as feeder loss and interface loss. All of system parameters are listed in Table II.

The dB-valued large-scale components such as path loss and shadow fading can be extracted from measurement data, which are given as follows:

$$L(d) = A + 10n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma \quad (2)$$

where A is intercept value and n is path loss exponent (PLE), both of which are obtained by using a Least Square (LS) regression fit. Parameter d_0 is reference distance where the value is 1 m. Parameter X_σ represents shadow fading component, which is dB-valued and is a zero-mean Gaussian distributed random variable with standard deviation σ .

Table III demonstrates complete large-scale characteristic parameters of horizontally and vertically polarized 1.8 GHz and 5.8 GHz propagating in four different tunnel scenarios. It should be pointed out that channel parameters in Table III are average values based on the results obtained from all five measurements. In order to eliminate near-field effect caused by antenna directivity, we ignore the data within first 50 m. Fig. 3 shows PL and LS fitting results for 1.8 GHz and 5.8 GHz in straight and curved tunnels and compares them with free-space model. Combined with Table III, the following four points can be discovered:

1) In straight tunnel scenarios, PLEs of horizontal and vertical polarization are 1.96, 2.11 (1.8 GHz) and 1.71, 1.98 (5.8 GHz), respectively. It can be seen from Fig. 3 that higher frequency leads to a larger path loss and smaller PLE which is significantly different from existing conclusions in outdoor open scenarios. Besides, for two different polarizations, path loss of H-Pol is slightly lower than that of V-Pol and its signal fluctuation amplitude is also smaller, which is strongly related to the cross-sectional dimension of subway tunnels. According to modal analysis theory, a larger tunnel width is more conducive to horizontal polarization, resulting in less propagation attenuation

TABLE III
LARGE-SCALE CHARACTERISTIC PARAMETERS IN SUBWAY TUNNELS AT
1.8 GHz AND 5.8 GHz FROM MEASUREMENTS

Freq.	Scenarios	Pol.	n	σ [dB]	d_{cor} [m]
1.8 GHz	Straight Tunnel	HH	1.96	1.12	5.22
		VV	2.11	2.81	6.11
	Straight Tunnel and Station	HH	1.99	1.90	6.01
		VV	2.33	2.19	6.18
	Curved Tunnel (R = 420 m)	HH	3.11	2.23	6.16
		VV	3.50	3.37	6.47
5.8 GHz	Straight Tunnel	HH	1.71	2.58	3.61
		VV	1.98	2.93	4.64
	Straight Tunnel and Station	HH	1.37	2.51	5.26
		VV	1.84	3.69	5.36
	Curved Tunnel (R = 420 m)	HH	1.92	2.42	3.38
		VV	2.39	2.76	5.60
5.8 GHz	Curved Tunnel (R = 890 m)	HH	1.57	2.52	5.03
		VV	1.79	2.53	5.92

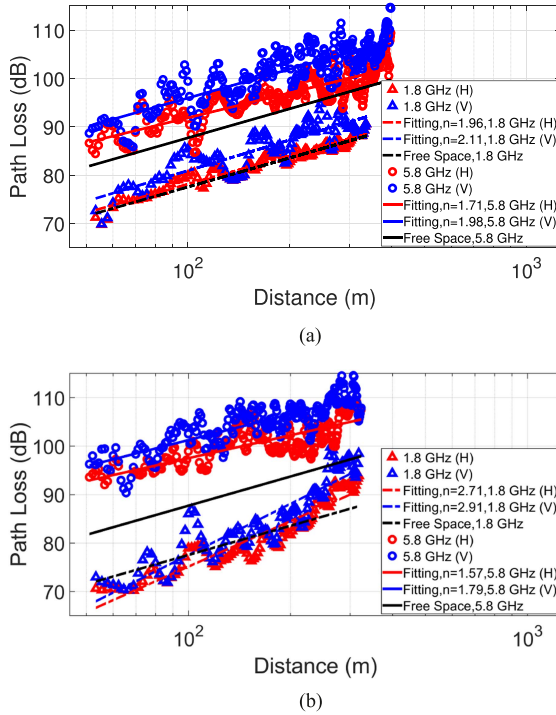


Fig. 3. Path Loss and LS fitting in: (a) Straight tunnel at 1.8 GHz and 5.8 GHz. (b) Curved tunnel (R = 890 m) at 1.8 GHz and 5.8 GHz.

and smoother fluctuation amplitude. Above conclusions are also reflected in [32], [34], [48].

2) In straight tunnel and station scenarios, PLEs of horizontal and vertical polarization are 1.99, 2.33 (1.8 GHz) and 1.37, 1.84 (5.8 GHz), respectively. Compared with straight tunnel, path loss at 1.8 GHz does not change much. However, path loss at 5.8 GHz shows lower propagation attenuation, especially for horizontally

polarized waves. This implies that stations have obvious impact on 5.8 GHz in terms of reducing propagation loss. Besides, path loss raised by inner wall of tunnels is also reduced, and path loss of horizontally polarized waves significantly declines.

3) In curved tunnel scenarios, when radius of curvature is 420 m, PLEs of horizontal and vertical polarization are 3.11, 3.50 (1.8 GHz) and 1.92, 2.39 (5.8 GHz) respectively, and as radius corresponding to 890 m, PLEs are 2.71, 2.91 (1.8 GHz) and 1.57, 1.79 (5.8 GHz) respectively. It can be found that, compared with straight tunnel, curved tunnel leads to additional path loss. Small radius of curvature leads to increased PLE since curved part of tunnel results in excessive reflection and refraction. Apart from these, in curved tunnels, horizontally polarized waves still attenuate less.

4) Compared with free-space model, it can be seen that 1.8 GHz achieves better proximity to free-space model than 5.8 GHz. Basing on propagation area division model [14], [27], 5.8 GHz signals mainly propagate in multi-mode propagation while 1.8 GHz signals fit free space propagation in measurement regions. Moreover, it can also be found that H-Pol transmits closer to free-space propagation than V-Pol because path loss of H-Pol is lower.

Probability density function (PDF) and cumulative probability function (CDF) of 1.8 GHz and 5.8 GHz shadow fading based on total measurements are shown in Fig. 4 for straight tunnel scenario as an example, and we summarize standard deviations of shadow fading for all tunnel scenarios in Table III, from which we can find that standard deviation of 1.8 GHz and 5.8 GHz is between 1.11–3.37 dB and 2.12–3.69 dB, respectively. Fig. 4 demonstrates that shadow fading in subway tunnels can be well fitted to a zero-mean Gaussian distribution, and fluctuation of 5.8 GHz is more severe than 1.8 GHz signals, which is consistent with results in Fig. 3. Moreover, standard deviation of 5.8 GHz in Table III is generally higher than 1.8 GHz.

In addition to frequency, polarization and curvature of tunnel mentioned above, it should also be pointed that dimension, cross-section, material of the tunnel will have different impacts on radio propagation, which has been confirmed by abundant relevant researches. Compared with the measurement results in straight road tunnel with 10 m width and 5 m height at 1.89 GHz [42], value of n is 1.68, which is smaller than 1.96–2.11 listed in Table III. However, values of n are 2.12 (H-Pol) and 2.46 (V-Pol) in straight subway tunnel with smaller dimension (3.43 m $\times$ 2.6 m) at 1.8 GHz [40]. Another example is that PLE n in curved tunnel with curvature of 500 m and dimension of 7.59 m $\times$ 5.52 m are 2.41 (H-Pol) and 2.20 (V-Pol) at 5705 MHz [48]. Thus it can be generally concluded that larger cross section dimension of tunnel leads to less attenuation. Moreover, roughness of tunnel is also found to affect large scale channel characteristics. For straight underground gallery with smaller cross section dimension and rougher walls, it has larger n and σ at 5.9 GHz, which are 2.48 and 3.80 dB respectively [54].

B. Autocorrelation of Shadow Fading

Shadow fading autocorrelation is defined as correlation in different positions of signals from the same Tx [55], which can

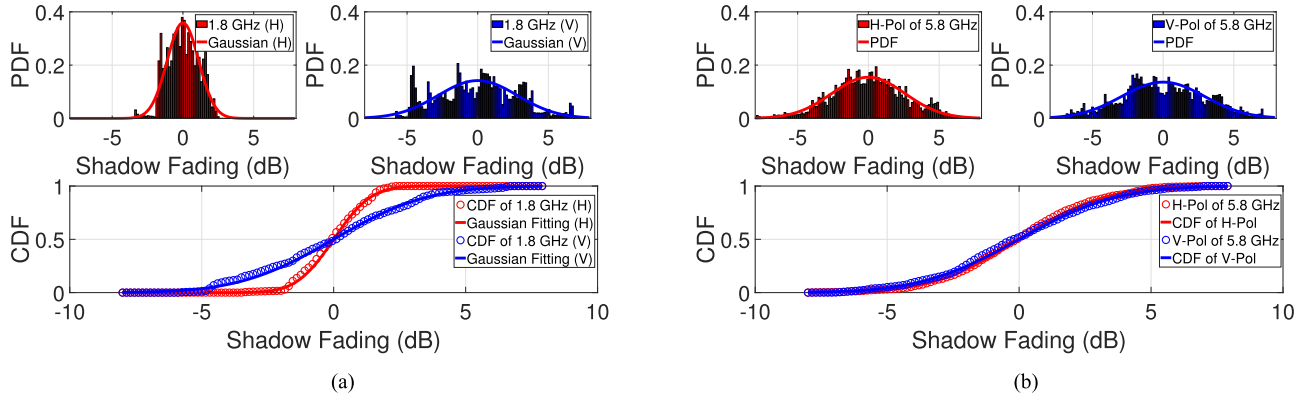


Fig. 4. PDFs and CDFs in straight tunnel at (a) 1.8 GHz. (b) 5.8 GHz.

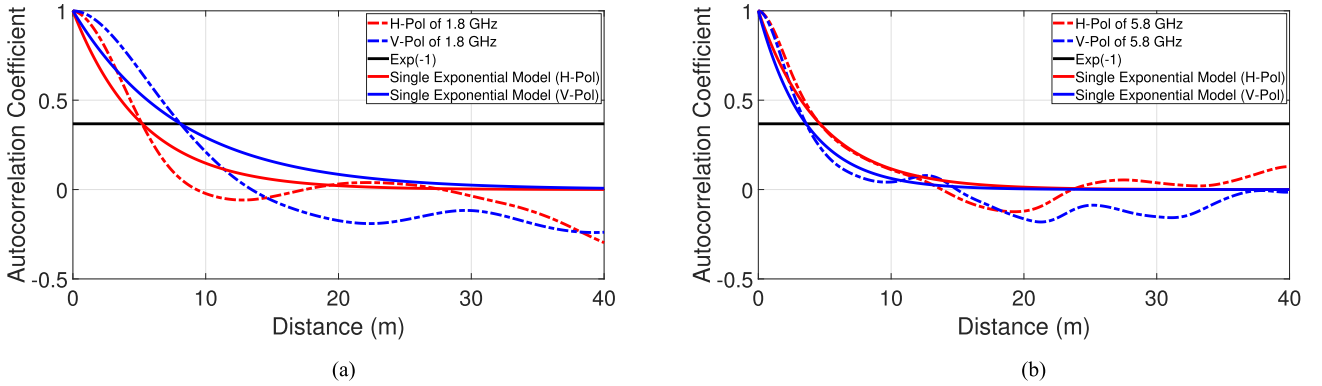


Fig. 5. Measurement and theoretical results by using Single Exponential Model fitting of autocorrelation coefficient of shadow fading in straight tunnel. (a) 1.8 GHz. (b) 5.8 GHz.

be expressed as

$$\rho_{auto}(\Delta d) = \frac{E\{X(d)X(d + \Delta d)\}}{\sigma_d \sigma_{d+\Delta d}} \quad (3)$$

where ρ_{auto} is autocorrelation coefficient, $X(d)$ and σ_d are values of shadow fading and standard deviation at position d , respectively. Parameter Δd represents the distance between two observed positions, and $E(\cdot)$ denotes mean value of set $(\cdot)$. Based on measurements, ρ_{auto} is found to follow a Single Exponential Model, expressed as

$$\rho_{auto}(\Delta d) = \exp\left(-\frac{\Delta d}{d_{cor}}\right) \quad (4)$$

where d_{cor} represents decorrelation distance, which is defined to be the distance at which ρ_{auto} drops to $1/e$. Fig. 5(a) and (b) demonstrate the measured and theoretical results at 1.8 GHz and 5.8 GHz in straight tunnel based on multiple measurements, respectively.

Decorrelation distances of shadow fading for all tunnel scenarios are shown in Table III. Based on multiple measurement results of all tunnel scenarios, we can find that measured data is fitted well by Single Exponential Model, especially for 5.8 GHz. Decorrelation distances of 1.8 GHz and 5.8 GHz are 5.22–8.49 m

and 3.38–5.92 m respectively, which demonstrates that higher frequency has smaller decorrelation distance of shadow fading in subway tunnels. Regardless of frequency, decorrelation distances of H-Pol and V-Pol are 3.38–8.02 m and 4.64–8.49 m, respectively. Compared with V-Pol, H-Pol has smaller decorrelation distances and faster change of shadow fading components. Decorrelation distances in four tunnel scenarios at 1.8 GHz are 5.22–6.11 m, 6.01–6.18 m, 6.16–6.47 and 8.02–8.49 m, respectively. Generally speaking, station and tunnel curvature will increase shadow fading decorrelation distance, and smaller curvature radius will bring smaller decorrelation distance. Situation of 5.8 GHz and 1.8 GHz is similar and is not repeated here. Decorrelation distance in subway tunnels is generally less than 10 m, which is significantly different from that in open outdoor scenarios, such as 57.12 m in urban, 112.48 m in suburban, and 114.79 m in rural at 930 MHz [55], mainly due to rich reflection and scattering components causing by walls around tunnels. Similar phenomenon is also reflected in [48], where decorrelation distances of curved tunnels with curvature of 300 m and 500 m are 3.23–5.50 m (V-Pol) and 1.79–5.78 m (H-Pol) at 5705 MHz, respectively. It can be implied that shadow fading components in tunnel change much faster along distance compared with other open area.

TABLE IV
SMALL-SCALE CHARACTERISTIC PARAMETERS IN SUBWAY TUNNEL AT 1.8 GHz AND 5.8 GHz FROM MEASUREMENTS

Freq.	Scenarios	FD (dB) [HH / VV]	Threshold (dB)	LCR (crossings/s) [HH / VV]	AFD (s) [HH / VV]
1.8 GHz	Straight Tunnel	50% -0.19 / -0.11	-5	0.82 / 0.41	0.120 / 0.010
		1% -4.38 / -6.27	0	27.88 / 23.34	0.023 / 0.038
		Δ 4.19 / 6.16	5	0.12 / 0.31	0.852 / 0.792
	Straight Tunnel and Station	50% -0.25 / -0.32	-5	13.43 / 15.73	0.009 / 0.012
		1% -5.38 / -7.10	0	24.65 / 21.56	0.031 / 0.035
		Δ 5.12 / 6.78	5	1.24 / 1.41	0.800 / 0.703
	Curved Tunnel (R = 420 m)	50% -0.31 / -0.43	-5	0.41 / 9.38	0.005 / 0.008
		1% -3.87 / -6.11	0	23.34 / 13.61	0.029 / 0.053
		Δ 3.57 / 5.68	5	0.31 / 1.43	3.266 / 0.696
	Curved Tunnel (R = 890 m)	50% -0.26 / -0.19	-5	1.59 / 12.70	0.006 / 0.007
		1% -4.35 / -6.17	0	38.97 / 29.41	0.017 / 0.025
		Δ 4.09 / 5.98	5	0.29 / 0.67	3.416 / 1.487
5.8 GHz	Straight Tunnel	50% -0.06 / -0.01	-5	0.96 / 1.73	0.015 / 0.010
		1% -5.18 / -5.65	0	71.15 / 48.75	0.009 / 0.014
		Δ 5.12 / 5.64	5	0.10 / 0.19	10.390 / 5.192
	Straight Tunnel and Station	50% -0.13 / -0.08	-5	2.24 / 2.30	0.003 / 0.005
		1% -3.80 / -4.26	0	67.00 / 37.95	0.010 / 0.019
		Δ 3.67 / 4.18	5	0.86 / 1.32	1.159 / 0.755
	Curved Tunnel (R = 420 m)	50% -0.11 / -0.42	-5	0.72 / 2.55	0.002 / 0.004
		1% -3.41 / -6.03	0	77.16 / 61.29	0.008 / 0.011
		Δ 3.32 / 5.60	5	0.82 / 0.76	1.224 / 1.313
	Curved Tunnel (R = 890 m)	50% -0.09 / -0.05	-5	0.94 / 2.08	0.004 / 0.003
		1% -3.51 / -4.01	0	69.50 / 45.00	0.009 / 0.016
		Δ 3.42 / 3.96	5	0.08 / 0.61	12.265 / 1.631

IV. RESULTS AND ANALYSIS OF SMALL-SCALE FADING

A. Fade Depth

Fade depth is used to measure fluctuation of channel energy relative to local mean value caused by small-scale fading, which can be defined as the difference between 50% level and 1% level of small-scale fading [56]. Table IV summarizes fading depth statistics of all measurement data in four types of subway tunnel scenarios. It can be found that FD does not change much for different scenarios and the maximum margin does not exceed 2 dB in general. FD of 5.8 GHz and H-pol is slightly lower than that of 1.8 GHz and V-pol respectively. The above results demonstrate that FD in tunnel is smaller than that in open scenarios due to waveguide effect, such as 14.38–15.96 dB in high-speed rail viaduct scenario [57] and 14.88–27.96 dB in crossing bridges scenario [58] at 930 MHz. The existence of stations has different effects on two frequencies. For 1.8 GHz, it increases FD and for 5.8 GHz, it causes FD fallen. However, influence brought by stations gives rise to data fluctuation about only 1–2 dB, which is relatively small. An interesting phenomenon is that curvature tends to reduce FD. In other words, tunnel curvature does not lead to more serious small-scale fading. Moreover, tunnels with greater curvature generally have smaller FD.

B. Level Crossing Rate and Average Fade Duration

LCR and AFD are second-order statistics to measure occurrence rate of burst errors and average length of burst errors. LCR is defined as average times the signal envelope positively passes a specified threshold level per second and AFD means average time that the signal envelope is below a certain threshold level [59]. It can lead to low signal-to-noise ratio, occurrence of burst errors and reduce performance if received signal envelope is less than a threshold. Therefore, LCR and AFD can be essential references for selections of transmission bit rate, frame length and encoding method in digital communication system design [60]. Since fluctuation of small-scale fading in subway tunnels is slight as mentioned above, we only set three threshold levels after normalizing signal envelopes of small-scale fading, i.e., -5 dB, 0 dB, and 5 dB, respectively. LCR and AFD of four subway tunnel scenarios are shown in Fig. 6. LCR and AFD of three threshold levels are given in Table IV. From Fig. 6 and Table IV, it can be seen that tunnel curvature increases LCR and existence of station does not have a significant impact. When threshold level is given, LCR of H-Pol is higher than that of V-Pol, and situation of AFD is the opposite, which demonstrates that small-scale envelope fluctuation of H-Pol is faster and more volatile in subway tunnels. Compared with 1.8 GHz, LCR of

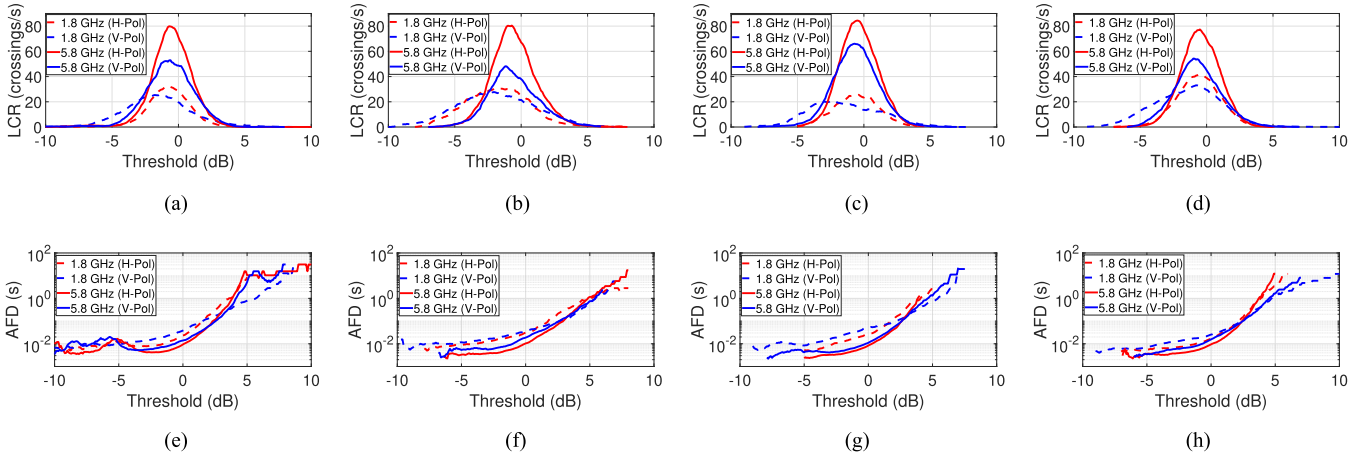


Fig. 6. (a) and (e) LCR and AFD in straight tunnel. (b) and (f) LCR and AFD in straight tunnel and station (c) and (g) LCR and AFD in curved tunnel ($R = 420$ m). (d) and (h) LCR and AFD in curved tunnel ($R = 890$ m).

5.8 GHz is higher and AFD is lower, which shows that fluctuation of small-scale of 5.8 GHz is more severe in subway tunnel. Similar results about LCR and AFD in tunnels are rarely found in the existing references. An example is that results of LCR and AFD with thresholds of 0, 5, 10 and 15 dB are calculated in straight road tunnel at 1.89 GHz [42], which are 0.72 crossings/s and 0.57 s, 0.28 s, 0.16 s, 0.11 s respectively. Also in the curved subway tunnels but 2.4 GHz [50], LCR and AFD with thresholds of 5 dB and 10 dB are 0.008λ , 0.001λ , and $0.098 \lambda^{-1}$, $0.007 \lambda^{-1}$, respectively.

C. Envelope Distribution

For analyzing envelope distribution of small-scale fading, we use non-overlapping sliding windows of 40 wavelengths on distance scale to divide small-scale fading into several observation intervals [61]. There is actually no direct path between Tx and Rx since receiving antennas are located inside subway locomotives and subway tunnels are narrow, as shown in Fig. 1. However, more abundant reflection and refraction paths can be generated in such environment. Consequently, the best fit distribution of amplitudes is entirely worthy of investigation. Four common and widely used candidate distributions are considered, namely Rician, Nakagami, Rayleigh and Weibull distributions. To select the model that best fits the empirical distribution of the fading amplitudes, Akaike's Information Criteria (AIC) is used. AIC is a decision method to measure the relative coincidence between statistical model and measured sample, which is widely used in the selection of small-scale fading distribution of wireless channels [62]. To streamline the comparison of AIC decision results, we opt for the approach described in [59], which entails selecting the distribution function with the highest AIC weight coefficient, signifying superior model fitting performance.

CDFs of distribution fitting and results of AIC weight coefficients among four candidate distributions at 5.8 GHz (H-Pol) in straight tunnel are shown in Fig. 7. In Fig. 7(a), except for Rayleigh distribution, the other candidate distributions all fit well, similar phenomenon also appears in [28] and [42]. However, Fig. 7(b) demonstrates that Nakagami distribution with

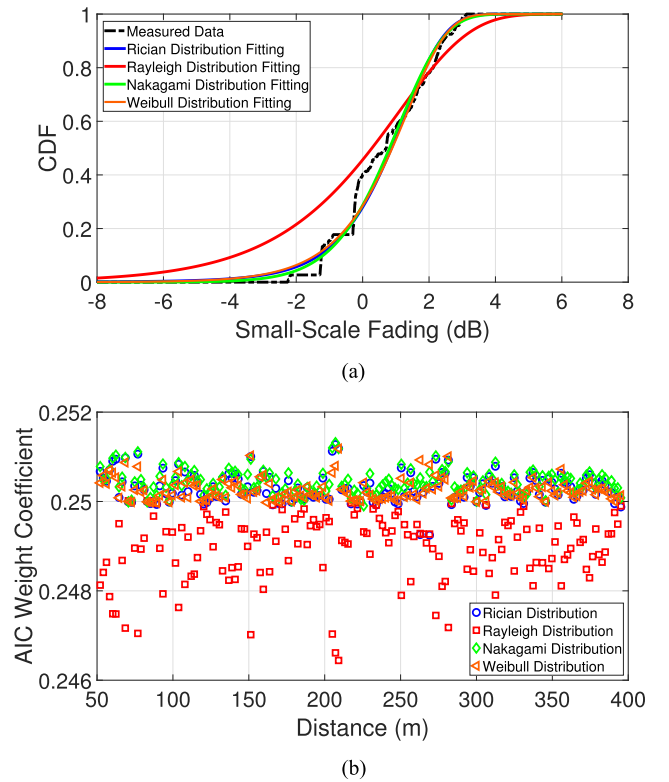


Fig. 7. (a) Small scale fading envelope distribution at 5.8 GHz (H-Pol) in Straight tunnel. (b) AIC weight coefficients and the percentage of the best fit for four candidate distributions at 5.8 GHz (H-Pol) in Straight tunnel.

the largest AIC weight coefficients is even better. Therefore, we consider Nakagami distribution as the best statistical distribution of small-scale fading envelope, which is also widely used in [42], [50], [63], [64] on small-scale statistical distribution in subway tunnels.

As a significant parameter to characterize fading, parameters- m of Nakagami distribution represents degree of fading and larger m signifies more severe fading [65], [66]. Parameter- m of Nakagami distribution in each measurement scenario and observation interval is extracted by a moment-based method and

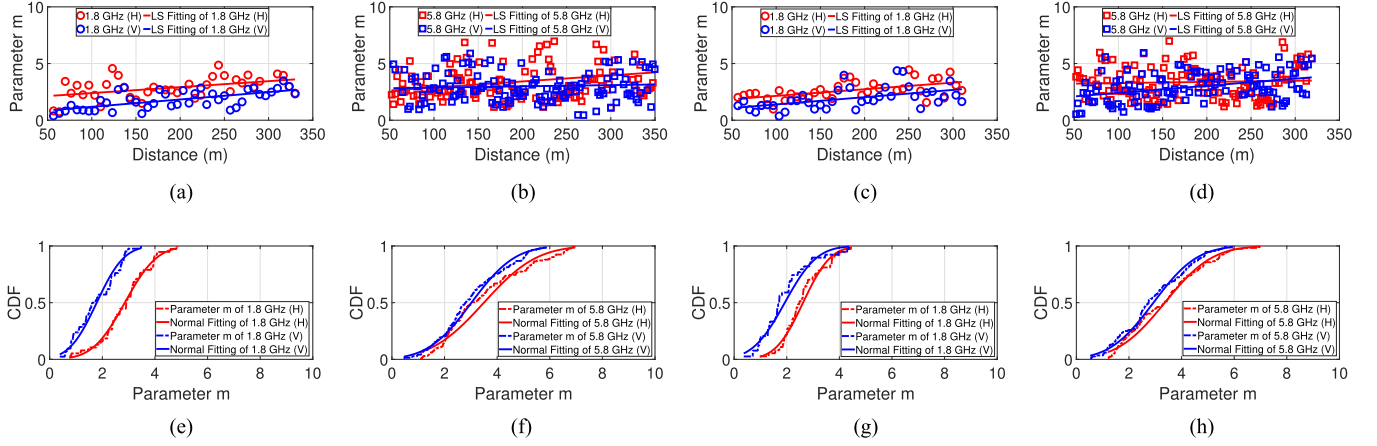


Fig. 8. Distribution and CDFs of parameter m . (a) and (e) 1.8 GHz in straight tunnel. (b) and (f) 5.8 GHz in straight tunnel. (c) and (g) 1.8 GHz in curved tunnel ($R = 890$ m). (d) and (h) 5.8 GHz in curved tunnel ($R = 890$ m).

TABLE V
PARAMETER- M IN TUNNELS AT 1.8 AND 5.8 GHz FROM MEASUREMENTS

Freq.	Scenarios	Pol.	m_1	m_2	Std
1.8 GHz	Straight Tunnel	HH	0.0053	1.8679	0.9621
		VV	0.0065	0.5312	0.8187
	Straight Tunnel and Station	HH	0.0015	1.0687	0.3220
		VV	0.0032	0.5418	0.3986
	Curved Tunnel ($R=420$ m)	HH	0.0137	0.6141	1.6144
		VV	0.0050	1.3946	1.3164
	Curved Tunnel ($R=890$ m)	HH	0.0059	1.5600	0.8213
		VV	0.0064	0.7541	0.9616
5.8 GHz	Straight Tunnel	HH	0.0011	2.3277	1.8845
		VV	0.0068	2.7690	1.2973
	Straight Tunnel and Station	HH	0.0018	6.8479	3.5300
		VV	0.0044	5.2686	3.3377
	Curved Tunnel ($R=420$ m)	HH	0.0031	3.1146	1.2388
		VV	0.0054	1.9186	1.2513
	Curved Tunnel ($R=890$ m)	HH	0.0033	3.3260	2.3891
		VV	0.0088	1.9228	3.5625

statistical distribution of m is analyzed in different tunnel scenarios. Taking straight and curved tunnels as examples, change of m with distance is shown in Fig. 8 and can be defined as:

$$m = m_1 d + m_2 \quad (5)$$

where m_1 and m_2 are tunable parameters which can be obtained by LS fitting.

Fig. 8 shows CDFs of m and Normal fitting distribution based on measurements, and tunable parameters and standard deviation of above model under all four different tunnel scenarios are listed in Table V. It can be found that Normal distribution can fit distribution of m well, which is positively correlated with distance between transceivers. At 1.8 GHz, m of H-Pol is higher than V-Pol, and this difference is not obvious at 5.8 GHz.

Parameter- m of 5.8 GHz is higher than that of 1.8 GHz, and more dispersed. Parameter- m is obviously higher in curved tunnels than straight tunnels, which shows that horizontally polarized and small-scale fading envelope of 5.8 GHz change more drastically, and tunnel curvature will aggravate such change.

V. CONCLUSION

In this paper, a set of comprehensive narrowband propagation measurements are conducted in realistic subway tunnels with four scenarios, which are straight tunnel, station, and curved tunnels with radii of curvature of 420 m and 890 m at 1.8 GHz and 5.8 GHz with H-Pol and V-Pol, respectively. Large and small-scale channel characteristics at 1.8 GHz and 5.8 GHz are investigated based on measurements, including PL, SF, channel correlation, FD, LCR, AFD and envelope distribution of small-scale fading. It can be found that frequency, polarization, tunnel curvature and station have significant impacts on wireless propagation. In general, higher frequency, V-Pol, tunnel curvature and station will lead to severe fading. Nakagami distribution fits envelope of small-scale fading well, and parameter- m increases with distance. Therefore, additional attention should be paid to these factors in design of subway communication systems. In future tunnel construction and planning, for example, when width of tunnel to be constructed is greater than height, horizontally polarized antennas should be selected as much as possible, and vice versa, vertically polarized antennas will be better. In the vicinity of curved tunnels and stations, transmitting power of nearby base stations should be increased or distance between base stations should be reduced to ensure sufficient signal strength. Results of AFD, LCR and Nakagami distribution can be used in channel simulation to study signal-to-noise ratio or bit-error rate characteristics in tunnels.

The qualitative analysis and corresponding results are expected to provide guidance for link budget, network planning, error control coding, interleaving, equalization algorithms, and validation for simulators of various wireless communication

systems deployed in real tunnels, which can further support development of future subway wireless communication systems.

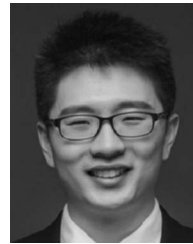
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Ruisi He (Senior Member, IEEE) received the B.E. and Ph.D. degrees from Beijing Jiaotong University (BJTU), Beijing, China, in 2009 and 2015, respectively. He is currently a Professor with the School of Electronics and Information Engineering, BJTU. He was a Visiting Scholar with the Georgia Institute of Technology, Atlanta, GA, USA, University of Southern California, Los Angeles, CA, USA, and Université Catholique de Louvain, Ottignies-Louvain-la-Neuve, Belgium. He has authored or coauthored eight books, five book chapters, more than 200 journal and conference papers, and several patents. His research interests include wireless propagation channels, 5G and 6G communications. He was an Editor of IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION, *IEEE Antennas and Propagation Magazine*, IEEE COMMUNICATIONS LETTERS, IEEE OPEN JOURNAL OF VEHICULAR TECHNOLOGY, and a Lead Guest Editor of IEEE JOURNAL ON SELECTED AREA IN COMMUNICATIONS and IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION. He was the Early Career Representative (ECR) of Commission C, International Union of Radio Science (URSI). Dr. He was the recipient of the URSI Issac Koga Gold Medal in 2021, IEEE ComSoc Asia-Pacific Outstanding Young Researcher Award in 2019, URSI Young Scientist Award in 2015, and several Best Paper awards in IEEE journals and conferences.



Mi Yang (Member, IEEE) received the M.S. and Ph.D. degrees from Beijing Jiaotong University, Beijing, China, in 2017 and 2021, respectively. He is currently an Associate Professor with the School of Electronic Information Engineering, Beijing Jiaotong University. His research interests include wireless channel measurement and modeling, vehicular and railway communications, and AI in channel research.



Ziyi Qi is currently working toward the Ph.D degree with School of Electronic Information Engineering, Beijing Jiaotong University, Beijing, China. His research interests include wireless channel measurement and modeling, wireless propagation environment reconstruction, and deterministic channel modeling.



Xuejian Zhang received the B.S. degree in communication engineering from Lanzhou Jiaotong University, Lanzhou, China, in 2021. He is currently working toward the Ph.D. degree with School of Electronic Information Engineering, Beijing Jiaotong University, Beijing, China. His research interests include wireless channel measurement and modeling, railway and vehicular communications, and computer vision aided wireless communication.



Zhengyu Zhang (Student Member, IEEE) received the B.S. and M.S. degrees in electronic and communication engineering in 2018 and 2021, respectively, from Beijing Jiaotong University, Beijing, China, where he is currently working toward the Ph.D. degree in information and communication systems. His research interests include ISAC, wireless communications, channel characterization and modeling, and deep learning.



Bo Ai (Fellow, IEEE) received the master's and Ph.D. degrees from Xidian University, Xi'an, China. He was a Visiting Professor with the Electrical Engineering Department, Stanford University, Stanford, CA, USA, in 2015. He is currently a Full Professor with Beijing Jiaotong University, Beijing, China, where he is also the Deputy Director with the State Key Laboratory of Advanced Rail Autonomous Operation, and the Deputy Director with International Joint Research Center. He is one of the Directors for Beijing

Science and Technology Cooperation Base, and the Backbone Member of the Innovative Engineering based jointly granted by the Chinese Ministry of Education and the State Administration of Foreign Experts Affairs. He has authored or coauthored eight books and authored more than 300 academic research papers in his research interests which include research and applications of channel measurement and channel modeling and dedicated mobile communications for rail traffic systems. Five papers have been the ESI highly cited paper and he also holds 26 invention patents. He is the Research Team Leader of 26 National Projects. He has also been notified by the Council of Canadian Academies that based on the Scopus database, and has been listed as one of the Top 1% authors in his field all over the world. He has been feature interviewed by the *IET Electronics Letters*. Dr. Ai is the Fellow of The Institution of Engineering and Technology and an IEEE VTS Distinguished Lecturer. He was the recipient of Distinguished Youth Foundation and Excellent Youth Foundation from the National Natural Science Foundation of China, Qiushi Outstanding Youth Award by the Hong Kong Qiushi Foundation, New Century Talents by the Chinese Ministry of Education, Zhan Tianyou Railway Science and Technology Award by the Chinese Ministry of Railways, Science and Technology New Star by the Beijing Municipal Science and Technology Commission, some important Scientific Research prizes, and the Honor of Excellent Postdoctoral Research Fellow from Tsinghua University, Beijing, China, in 2007. He is an IEEE VTS Beijing Chapter Vice Chair and an IEEE BTS Xi'an Chapter Chair. He was the Co-Chair or a Session/Track Chair of many international conferences. He is an Associate Editor for IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS and IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, and the Editorial Committee Member of *Wireless Personal Communications Journal*. He is a Lead Guest Editor of special issues on IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS, and *International Journal on Antennas and Propagation*.



Ruifeng Chen received the B.E. and Ph.D. degrees from Beijing Jiaotong University, Beijing, China, in 2010 and 2016, respectively. Since 2015, she is currently with the Institute of Computing Technologies, China Academy of Railway Sciences Corporation Limited, Beijing, China. She has been a Visiting Scholar with the University of British Columbia, Vancouver, BC, Canada. She was the recipient of the Young Talent Sponsorship Program of China Association for Science and Technology. Her research interests include Internet of Things, wireless sensor networks, and railway 5G communications and channels.